

Kiran Ilyas

RF & Electromagnetic Simulation Engineer | AI/ML-Assisted Modeling | Electronics Engineer

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PROFESSIONAL SUMMARY

Engineering Doctorate (EngD)-qualified RF and Electromagnetic Simulation Engineer with industrial R&D experience at NXP Semiconductors, with specialization in mm-wave RFIC electromagnetic analysis, radiated emission characterization, and cross-software simulation benchmarking using Finite Element Method (FEM) in ANSYS HFSS and both FEM and Method of Moments (MoM) in Keysight ADS. Managed the full project lifecycle from problem definition and simulation methodology scoping through execution, milestone tracking, and formal deliverable reporting within an EU-funded industrial research consortium. Developed an AI/ML-assisted near-field to far-field (NF-FF) prediction pipeline in PyTorch, demonstrated as a computationally efficient alternative to full-wave EM simulation. Strong publication record and direct references from NXP Principal Scientist and TU/e faculty.

TECHNICAL SKILLS

EM & RF Simulation: ANSYS HFSS (FEM), Keysight ADS (FEM/MoM), CST Studio Suite, COMSOL Multiphysics, ANSYS optiSLang (DOE)

EDA: Cadence Virtuoso

AI/ML & Computing: PyTorch (CNN, MLP, U-Net) · Python (NumPy, Jupyter Notebook) · MATLAB · C · LaTeX

OS & Environment: Linux (HPC remote servers, TU/e & NXP environments) · Windows

Domains: RF/mm-Wave Engineering (100 GHz) · Electromagnetic Compatibility (EMC) & EMI · Radiated Emission Characterization · NF-FF Modeling · Radiation Pattern Analysis · Multilayer Stackup Modelling · Semiconductor Technologies · Integrated Photonics

Methods: FEM · MoM · DOE & Parametric Simulation · Cross-Solver Benchmarking · Verification & Validation (V&V) · Simulation-Based ML · Iterative Model Development · End-to-End Project Lifecycle Management · Knowledge Sharing & Technical Communication

PROFESSIONAL EXPERIENCE

R&D Engineer – Computational Electromagnetics & AI/ML Modeling

NXP Semiconductors / Eindhoven University of Technology | Mar 2024 – Dec 2025

EU-Funded INNOSTAR Project Consortium | European 5G/6G Connectivity and Radar Applications

- Managed the full end-to-end project lifecycle: defined the research scope, simulation methodology, and ML approach in alignment with academic and industrial supervisors; executed all technical work packages; tracked milestone progress; and delivered formal project reviews, consortium reports, and an NXP-approved publication manuscript.
- Benchmarked fundamental mm-wave small radiating structures at 100 GHz by performing Near-Field (NF) and Far-Field (FF) radiated emission simulations in ANSYS HFSS (FEM) and Keysight ADS (FEM and MoM), systematically comparing radiation patterns and field results across solvers to assess consistency, solver behaviour, and S-parameter characteristics for high-frequency RFIC analysis across a frequency range of 10-100 GHz.
- Developed a requirements-driven NF-to-FF prediction workflow to address a project-defined measurement gap: near-field data from integrated circuits was available but corresponding far-field radiation data was not directly accessible. The resulting AI/ML pipeline predicts far-field radiation from near-field inputs, supporting radiated emission analysis and Electromagnetic Compatibility (EMC) verification workflows.
- Designed and executed a parametric Design of Experiments (DOE) study using ANSYS optiSLang on a two-layer stackup structure: varied geometrical dimensions at 100 GHz to produce a geometry-diverse set of NF and FF simulation datasets for machine-learning model development.
- Built Python (NumPy, Jupyter Notebook) data preprocessing pipelines to structure large-scale NF/FF simulation datasets into clean input-output pairs, applying iterative development and validation cycles to improve pipeline robustness and model training quality.
- Developed and evaluated CNN and U-Net deep-learning architectures in PyTorch for NF-to-FF field prediction, conducting systematic Verification & Validation (V&V) by benchmarking model predictions against full-wave simulation references across multiple loss functions.
- Actively shared technical knowledge and development progress with team members throughout the project, presenting methodology updates, intermediate results, and simulation findings to both academic and industrial colleagues to support team understanding of a novel computational workflow.

Project Team Coordinator

VDL Kalima, Eindhoven | Nov 2024 – Feb 2025

- Coordinated a multidisciplinary engineering project team, managing planning, milestone tracking, task coordination, component ordering, budget monitoring, and stakeholder communication within an industrial EngD environment.

Researcher – Nonlinear Optics & Integrated Photonics

Quaid-i-Azam University, Pakistan | 2021 – 2023

- Designed, characterized, and optimized dispersion-engineered lithium niobate and silicon-rich nitride photonic integrated waveguides for broadband supercontinuum generation across mid- and far-IR spectral regions.
- Modelled electromagnetic wave propagation, nonlinear pulse dynamics, and dispersive-wave generation using COMSOL Multiphysics and MATLAB, resulting in peer-reviewed journal and conference publications.

EDUCATION

Engineering Doctorate (EngD) in Electrical Engineering

Eindhoven University of Technology, Netherlands | 2024 – 2026

Graduated June 2026 | Thesis: AI/ML-Based Near-Field to Far-Field Mapping for mm-Wave ICs in EMC Modeling

Design & Knowledge: Design Methods for ICT-Systems · Statistics I & II (Practical Data Analysis, DOE) · Systems Thinking I & II · Systems Engineering Advanced (incl. SysML).

Electives: Numerical Methods in Electrical Engineering · Advanced CMOS Design

Master of Philosophy (M.Phil.) in Electronics

Quaid-i-Azam University, Pakistan | 2019 – 2021

CGPA: 4.0/4.0 — Gold Medalist | Thesis: Dispersion Engineered Mid-IR Supercontinuum Generation Using Dual-Core Silicon-Rich Nitride Photonic Integrated Waveguide

Coursework: Wave Propagation · Microwave Engineering · High-Frequency Electromagnetics · Radar Signal Processing & System Design · Advanced Digital Signal Processing · Electromagnetic Metamaterials · Microelectronics-I · Oversampling Data Converters

Master of Science (M.Sc.) in Electronics

Quaid-i-Azam University, Pakistan | 2017 – 2019

Ranked 3rd in Graduating Class

Mathematics: Applied Mathematics I & II

Electronics: Electronics I, II & III · Quantum & Physical Electronics · Network Analysis

Electromagnetics & Communications: Electromagnetic Theory · Communication Theory · Communication Systems · Wireless Communication

Systems, Programming & Design: Control Systems · Computer Architecture · Microprocessor-Based Design · Computer Programming · Digital Logic Design · Verilog-HDL

CERTIFICATIONS & TRAINING

- Machine Learning with MATLAB — MathWorks MATLAB Academy (Online)
- HPC Supercomputing Autumn School — High-performance computing for engineering simulation workloads
- Participation EMC Europe 2024 (Bruges) & EMC Europe 2025 (Paris) — International EMC conference attendance

PUBLICATIONS

- Ilyas, K. et al. "Reactive Near-Field to Far-Field Prediction in mm-Wave Integrated Circuits Using Deep Learning Architectures for EMC Modeling." Manuscript prepared for journal submission (NXP-approved).
- Ilyas, K., & Naqvi, Q. A. (2024). "Three-octave supercontinuum spanning with Z-cut lithium niobate photonic waveguide." Journal of Electromagnetic Waves and Applications.
- Ilyas, K., & Naqvi, Q. A. (2022). "Dispersion engineered mid-IR supercontinuum generation using multi-core silicon-rich nitride photonic integrated waveguide." Optik.
- Ilyas, K., & Naqvi, Q. A. (2022). "Tunable Dispersive Wave Generation for High Power Supercontinuum Coverage in Mid-IR and Far-IR Regions." Optica Advanced Photonics Congress.

LANGUAGES

English: C1 — Fluent (professional and academic) **Dutch:** A1 — Beginner (actively developing towards A2) **Urdu / Punjabi:** Native

AWARDS & HONOURS

- Chancellor's Medal — presented by the President of Pakistan for 1st position in the Faculty of Science, Quaid-i-Azam University (2023).
- Vice Chancellor's Medal — presented by the Federal Minister for Education for 1st position in the Department of Electronics, Quaid-i-Azam University (2023).
- Merit Scholarships throughout M.Sc. and M.Phil. studies based on academic excellence.

LEADERSHIP & PROFESSIONAL ACTIVITIES

- Event Lead — IEEE Women in Engineering, EMC Europe 2025: Organized an international networking event alongside the EMC Europe conference, managing sponsors, registrations, logistics, and programme.
- Student Mentor — Netherlands–Korea Semiconductor Technology Winter Academy, TU/e: Guided Korean university students through the academic program, laboratory visits, and university environment.
- President — SPIE Student Chapter, Quaid-i-Azam University: Organized technical workshops on COMSOL and Nonlinear Optics; managed speakers, budgets, logistics, and coordination.

REFERENCES

Available upon request. Direct references from NXP Semiconductors (Principal Scientist, Principal Engineer – EMC) and TU/e (Dept. of Electrical Engineering).